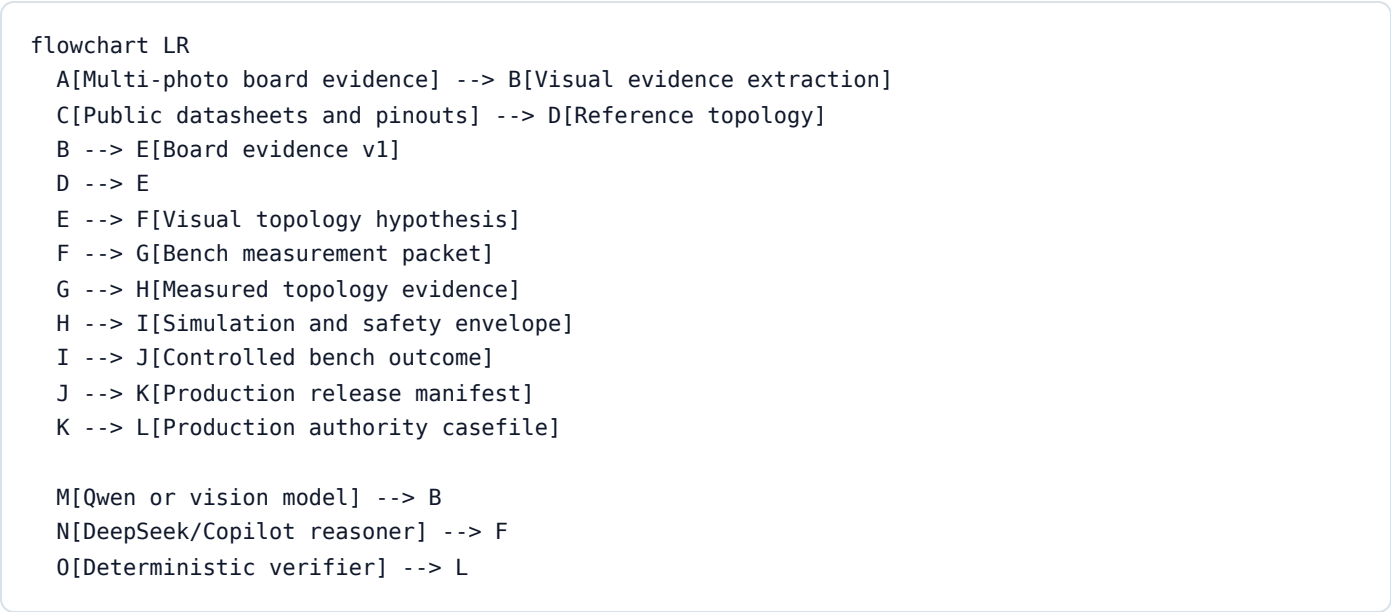


Circuit.AI Hardware Splicer Architecture

Core Claim

Circuit.AI is an evidence-gated AI agent for circuit board understanding, repair, reuse, and hardware recomposition. It does not treat visual model output as electrical truth. It converts model output into measurement plans and only promotes a board to repair authority after measured evidence and bench outcomes.

Architecture



Evidence Authority Levels

Level	Meaning	Allowed Output
visual_candidate	photos and reference hints only	capture plan and measurement queue
measured_topology	trusted pinout/rail measurements exist	scoped topology and simulation checks

Level	Meaning	Allowed Output
electrical_simulation	measured envelope passes deterministic checks	bench protocol and guarded bring-up
controlled_bench	first-power/function proof recorded	controlled reuse claim
production_repair	release manifest and artifacts complete	scoped production repair/reuse authority

Current Live Demo

Frontend:

```
/showcase?state=release
```

Backend:

```
POST /hardware/production-casefile/run
```

The showcase demonstrates:

- evidence packet selection
- board topology visualization
- measurement closure
- authority ladder
- live production casefile response
- runtime model/budget safety strip

Model Strategy

Models help interpret and plan:

- Qwen VL: candidate board evidence from multi-photo input.
- DeepSeek/Copilot-style reasoners: structured advisory reasoning.
- Local deterministic verifier: final authority gate.

The system is designed so model output can accelerate work but cannot authorize

unsafe hardware actions alone.

Competition Deliverable

The final demo will show a CH340C USB-serial board moving through:

```
reference-only evidence -> blocked authority  
measured topology -> simulation-ready authority  
release evidence -> production repair authorized
```